

Crypto Prediction Suite with High-Visibility Mobile HUD

```
<!DOCTYPE html>
<html lang="en" class="h-full bg-slate-950 text-slate-100">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
initial-scale=1.0">
  <title>Crypto Prediction Probability Engine &
Backtester</title>
  <!-- Tailwind CSS CDN (Whitelisted Domain) -->
  <script src="https://cdn.tailwindcss.com"></script>
  <style>
    @import
url('https://fonts.googleapis.com/css2?family=JetBrains+Mono:wght@4
00;500;700&family=Plus+Jakarta+Sans:wght@400;500;600;700&display=sw
ap');
```

```
    body {
      font-family: 'Plus Jakarta Sans', sans-serif;
    }
    .mono {
      font-family: 'JetBrains Mono', monospace;
    }
    /* Custom scrollbar styling */
    ::-webkit-scrollbar {
```

```
      width: 6px;
      height: 6px;
    }
    ::-webkit-scrollbar-track {
      background: #020617;
    }
    ::-webkit-scrollbar-thumb {
      background: #334155;
      border-radius: 3px;
    }
    ::-webkit-scrollbar-thumb:hover {
      background: #475569;
```

```
}
```

```
/* Animated Flow Line Effects */  
@keyframes flowProgress {  
  from { stroke-dashoffset: 24; }  
  to { stroke-dashoffset: 0; }  
}
```

```
.flow-line-active {  
  stroke-dasharray: 6, 4;  
  animation: flowProgress 1s linear infinite;  
}  
.flow-line-degraded {  
  stroke-dasharray: 4, 6;  
  animation: flowProgress 3s linear infinite;  
}  
</style>
```

```
</head>
```

```
<body class="h-full flex flex-col antialiased  
selection:bg-indigo-500/30">
```

```
<!-- Top Navigation Header -->  
<header class="border-b border-slate-800 bg-slate-900/50  
backdrop-blur-md px-6 py-4 flex flex-wrap items-center  
justify-between gap-4 sticky top-0 z-50">  
  <div class="flex items-center space-x-3">  
    <div class="h-9 w-9 rounded-xl bg-gradient-to-tr  
from-indigo-600 to-violet-500 flex items-center justify-center p-2  
shadow-lg shadow-indigo-500/20">  
      <svg class="w-5 h-5 text-white" fill="none"  
viewBox="0 0 24 24" stroke="currentColor">  
        <path stroke-linecap="round"  
stroke-linejoin="round" stroke-width="2.5" d="M13 7h8m0 0v8m0-8l-8  
8-4-4-6 6" />  
      </svg>
```

```
    </div>  
    <div>  
      <h1 class="font-bold text-lg tracking-tight  
text-white flex items-center gap-2">  
        PROB-ENGINE <span class="text-xs  
bg-indigo-500/10 text-indigo-400 font-mono font-medium px-2 py-0.5
```

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rounded border border-indigo-500/20">v3.8</span>
    </h1>
    <p class="text-xs text-slate-400">Continuous
15-Minute Crypto Prediction Matrix</p>
  </div>
</div>

<!-- Tab View Controls Suite -->
<div class="flex bg-slate-950 p-1 rounded-xl border
border-slate-800">
  <button id="tabLiveBtn"
onclick="switchViewMode('live')" class="px-4 py-1.5 text-xs
font-semibold rounded-lg bg-indigo-600 text-white transition
duration-150">
    Live Dashboard
  </button>
  <button id="tabBacktestBtn"
onclick="switchViewMode('backtest')" class="px-4 py-1.5 text-xs
font-semibold rounded-lg text-slate-400 hover:text-slate-200
transition duration-150">
    Bulk Backtest Matrix
  </button>
</div>

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<div class="flex items-center gap-6">
  <div class="text-right hidden sm:block">
    <p class="text-xs text-slate-400">System Time</p>
    <p id="systemClock" class="text-sm font-semibold
font-mono text-slate-200">00:00:00 UTC</p>
  </div>
  <div class="flex items-center gap-2 bg-emerald-500/10
border border-emerald-500/20 text-emerald-400 px-3 py-1.5
rounded-lg text-xs font-medium">
    <span class="relative flex h-2 w-2">
      <span class="animate-ping absolute inline-flex
h-full w-full rounded-full bg-emerald-400 opacity-75"></span>
      <span class="relative inline-flex rounded-full
h-2 w-2 bg-emerald-500"></span>
    </span>
    LIVE CORE ENGINE ACTIVE
  </div>
</div>
</header>

```

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    <!-- TAB CONTAINER ONE: LIVE PANEL LAYOUT WORKSPACE -->
    <main id="liveWorkspaceView" class="flex-1 max-w-[1600px]
w-full mx-auto p-4 lg:p-6 grid grid-cols-1 xl:grid-cols-12 gap-6
overflow-y-auto">

        <!-- Left Column: Inputs & Variables Configuration (4
Columns Wide) -->

        <section class="xl:col-span-4 flex flex-col gap-6">

            <!-- Card 1: Historical Sequence Setup -->
            <div class="bg-slate-900 border border-slate-800
rounded-2xl p-5 shadow-sm flex flex-col gap-4">
                <div class="flex items-center justify-between
border-b border-slate-800 pb-3">
                    <div class="flex items-center gap-2">
                        <svg class="w-5 h-5 text-indigo-400"
fill="none" viewBox="0 0 24 24" stroke="currentColor">
                            <path stroke-linecap="round"
stroke-linejoin="round" stroke-width="2" d="M12 8v4l3 3m6-3a9 9 0
11-18 0 9 9 0 0118 0z" />
                        </svg>
                        <h2 class="font-semibold
text-slate-200">Historical Cycles (Last 4-6 Results)</h2>
                    </div>
                    <span class="text-xs font-mono text-slate-400"
id="historicalCountLabel">6 Blocks</span>
                </div>

                <p class="text-xs text-slate-400 leading-relaxed">
                    Set the previous 15-minute predictive block
outcomes. Modern parameters apply greater decay weight to the most
recent cycles.
                </p>

            <!-- Interactive Blocks Container -->

            <div class="grid grid-cols-3 gap-2 my-2"
id="historicalBlocksGrid">
                <!-- Dynamic Blocks generated by Javascript -->
            </div>

```

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        <div class="flex items-center justify-between
pt-2">
            <span class="text-xs text-slate-400">Sample
Dataset Presets:</span>
            <div class="flex gap-2">
                <button
onclick="applyPreset([1,1,0,1,0,1])" class="text-[11px]
bg-slate-800 hover:bg-slate-700 text-slate-300 font-mono px-2 py-1
rounded transition">Bullish Bias</button>
                <button
onclick="applyPreset([0,0,1,0,1,0])" class="text-[11px]
bg-slate-800 hover:bg-slate-700 text-slate-300 font-mono px-2 py-1
rounded transition">Bearish Bias</button>
            </div>
        </div>
    </div>
</div>

```

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        <!-- Card 2: Strategy Threshold & Execution Pad -->
        <div class="bg-slate-900 border border-slate-800
rounded-2xl p-5 shadow-sm flex flex-col gap-4">
            <div class="flex items-center gap-2 border-b
border-slate-800 pb-3">
                <svg class="w-5 h-5 text-violet-400"
fill="none" viewBox="0 0 24 24" stroke="currentColor">
                    <path stroke-linecap="round"
stroke-linejoin="round" stroke-width="2" d="M12 6V4m0 2a2 2 0 100
4m0-4a2 2 0 110 4m-6 8a2 2 0 100-4m0 4a2 2 0 110-4m0 4v2m0-6V4m6
6v10m6-2a2 2 0 100-4m0 4a2 2 0 110-4m0 4v2m0-6V4" />
                </svg>
            </div>
        </div>

```

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        <h3 class="font-semibold
text-slate-200">Execution & Sentiment Tuning</h3>
    </div>

```

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        <!-- Probability Threshold Control -->
        <div class="flex flex-col gap-1.5">
            <div class="flex justify-between items-center
text-xs">
                <label for="thresholdSlider"
class="text-slate-300 font-medium">Confidence Barrier (KEEP Signal
Threshold)</label>
                <span id="thresholdValueDisplay"
class="font-mono text-indigo-400 font-semibold bg-indigo-500/10
px-1.5 py-0.5 rounded">55%</span>
            </div>
        </div>

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        <input type="range" id="thresholdSlider"
min="50" max="80" value="55" class="w-full h-1.5 bg-slate-800
rounded-lg appearance-none cursor-pointer accent-indigo-500">
    </div>

```

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    <!-- Intraday Micro Momentum Control -->
    <div class="flex flex-col gap-1.5">
        <div class="flex justify-between items-center
text-xs">
            <label for="momentumSlider"
class="text-slate-300 font-medium">Intra-Candle Ticker
Micro-Momentum</label>
            <span id="momentumValueDisplay"
class="font-mono text-emerald-400 font-semibold bg-emerald-500/10
px-1.5 py-0.5 rounded">+20%</span>
        </div>
        <input type="range" id="momentumSlider"
min="-50" max="50" value="20" class="w-full h-1.5 bg-slate-800
rounded-lg appearance-none cursor-pointer accent-emerald-500">

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    </div>

```

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    <!-- User Initial Intent Stance -->
    <div class="flex flex-col gap-2 border-t
border-slate-800/60 pt-3">
        <span class="text-xs font-medium
text-slate-300">Your Target Intent Stance</span>
        <div class="grid grid-cols-2 gap-3">
            <button id="stanceOverBtn"
onclick="setUserStance('OVER')" class="py-2 px-3 rounded-xl border
text-xs font-semibold tracking-wide font-mono transition flex
items-center justify-center gap-1.5 bg-emerald-500/10
border-emerald-500/40 text-emerald-400 shadow-sm
shadow-emerald-500/5">
                <span class="h-2 w-2 rounded-full
bg-emerald-400"></span> OVER
            </button>
            <button id="stanceUnderBtn"
onclick="setUserStance('UNDER')" class="py-2 px-3 rounded-xl border
text-xs font-semibold tracking-wide font-mono transition flex
items-center justify-center gap-1.5 bg-slate-800/40
border-slate-700 text-slate-400">
                <span class="h-2 w-2 rounded-full
bg-slate-500"></span> UNDER
            </button>

```

```

        </div>
    </div>

    <!-- Quick Bid Execution Pad -->
    <div class="flex flex-col gap-2 border-t
border-slate-800/60 pt-3">
        <div class="flex justify-between items-center">
            <span class="text-xs font-medium
text-slate-300">Quick Bid Size Layout</span>

            <span id="activeBidDisplay"
class="font-mono text-[11px] text-amber-400 font-bold
bg-amber-500/10 px-1.5 py-0.5 rounded">0.75 ETH</span>
        </div>
        <div class="grid grid-cols-4 gap-1.5">
            <button onclick="setQuickBid(0.25)"
class="py-1.5 text-xs font-mono font-bold rounded-lg border
border-slate-700 bg-slate-800/40 hover:bg-slate-800 text-slate-300
transition">.25</button>
            <button onclick="setQuickBid(0.50)"
class="py-1.5 text-xs font-mono font-bold rounded-lg border
border-slate-700 bg-slate-800/40 hover:bg-slate-800 text-slate-300
transition">.50</button>
            <button onclick="setQuickBid(0.75)"
class="py-1.5 text-xs font-mono font-bold rounded-lg border
border-indigo-500/40 bg-indigo-500/10 hover:bg-indigo-500/20
text-indigo-400 transition shadow-sm">.75</button>
            <button onclick="setQuickBid(1.00)"
class="py-1.5 text-xs font-mono font-bold rounded-lg border
border-slate-700 bg-slate-800/40 hover:bg-slate-800 text-slate-300
transition">1.0</button>
        </div>
        <div class="relative mt-1">
            <input type="number" id="manualBidInput"
value="0.75" step="0.01" min="0.01"
oninput="handleManualBidInput(this.value)" class="w-full
bg-slate-950 border border-slate-800 rounded-xl py-2 pl-3 pr-20
text-xs font-mono text-slate-100 focus:outline-none
focus:border-indigo-500">
            <span class="absolute right-3 top-2.5
text-[10px] uppercase font-mono font-bold text-slate-500
tracking-wider">ETH Unit</span>
        </div>

    <!-- Transmit Action Handler (Circuit Breaker

```

Managed) -->

```
        <button id="transmitBidActionBtn"
onclick="executeBidTransmissionSim()" class="w-full mt-1.5
bg-gradient-to-r from-amber-500 to-orange-600 hover:from-amber-600
hover:to-orange-700 text-slate-950 font-extrabold py-2.5 px-4
rounded-xl transition duration-150 flex items-center justify-center
gap-2 tracking-wide shadow-md shadow-orange-500/10
active:scale-[0.98]">
 TRANSMIT BID NOW (0.75 ETH)
        </button>
    </div>
</div>
```

```
    <!-- Card 3: Live Order Flow Tape -->
    <div class="bg-slate-900 border border-slate-800
rounded-2xl p-5 shadow-sm flex flex-col gap-3 h-[210px]">
        <div class="flex items-center justify-between
border-b border-slate-800 pb-2">
            <div class="flex items-center gap-2">
                <svg class="w-4 h-4 text-emerald-400
animate-pulse" fill="none" viewBox="0 0 24 24"
stroke="currentColor">
                    <path stroke-linecap="round"
stroke-linejoin="round" stroke-width="2" d="M9 12l2 4-4M7.835
4.697a3.42 3.42 0 001.946-.806 3.42 3.42 0 014.438 0 3.42 3.42 0
00.806 1.946 3.42 3.42 0 013.138 3.138 3.42 3.42 0 00.806 1.946
3.42 3.42 0 010 4.438 3.42 3.42 0 00-.806 1.946 3.42 3.42 0
01-3.138 3.138 3.42 3.42 0 00-1.946.806 3.42 3.42 0 01-4.438 0 3.42
3.42 0 00-1.946-.806 3.42 3.42 0 01-3.138-3.138 3.42 3.42 0
00-.806-1.946 3.42 3.42 0 010 4.438 3.42 3.42 0 00.806-1.946 3.42
3.42 0 013.138-3.138z" />
                </svg>
                <h3 class="font-semibold text-slate-200
text-sm">Live Order Flow Tape</h3>
            </div>
            <span class="text-[10px] font-mono
text-slate-500 uppercase tracking-wider">Order Book Stream</span>
        </div>
        <div id="tickerTapeStream" class="flex-1
overflow-y-auto bg-slate-950 rounded-xl border border-slate-800/80
p-3 font-mono text-[11px] flex flex-col gap-1">
            <!-- Flowing numbers are continuously inserted
here via JS -->
        </div>
```

```
    </div>
  </section>
```

```
    <!-- Center Column: Simulator, Flow Chart & Real-Time
    Probability Gauges (5 Columns Wide) -->
    <section class="xl:col-span-5 flex flex-col gap-6">
```

```
      <!-- Real-Time Timeline Control Deck -->
      <div class="bg-slate-900 border border-slate-800
rounded-2xl p-5 shadow-sm flex flex-col gap-4">
        <div class="flex items-center justify-between
border-b border-slate-800 pb-3">
          <div class="flex items-center gap-2">
            <svg class="w-5 h-5 text-amber-400"
fill="none" viewBox="0 0 24 24" stroke="currentColor">
              <path stroke-linecap="round"
stroke-linejoin="round" stroke-width="2" d="M11 3.055A9.001 9.001 0
1020.945 13H11V3.055z" />
              <path stroke-linecap="round"
stroke-linejoin="round" stroke-width="2" d="M20.488
9H15V3.512A9.025 9.025 0 0120.488 9z" />
            </svg>
            <h2 class="font-semibold
text-slate-200">15-Minute Candle Tracking Deck</h2>
          </div>
          <div class="flex items-center gap-1
bg-slate-950 rounded-lg p-0.5 border border-slate-800 font-mono
text-[11px]">
            <button id="modeSimBtn"
onclick="setTimerMode('sim')" class="px-2 py-1 rounded bg-slate-800
text-white font-medium">SIM</button>
            <button id="modeLiveBtn"
onclick="setTimerMode('live')" class="px-2 py-1 rounded
text-slate-400 hover:text-slate-200">LIVE COLD</button>
          </div>
        </div>
      </div>
```

```
      <!-- Simulation Controls Container -->
      <div class="grid grid-cols-1 md:grid-cols-2 gap-4
items-center bg-slate-950 p-4 rounded-xl border
border-slate-800/80">
```

```

        <div class="flex flex-col">
          <span class="text-[11px] text-slate-400
tracking-wider font-mono">TIME REMAINING</span>
          <span id="countdownTimerDisplay"
class="text-4xl font-bold font-mono tracking-tight text-white
my-0.5">15:00</span>
          <span id="elapsedSecondsDisplay"
class="text-[11px] font-mono text-slate-500">0s elapsed of
900s</span>
        </div>
        <div class="flex flex-col gap-2">
          <div class="flex gap-2 justify-end">
            <button id="playPauseBtn"
onclick="togglePlayback()" class="bg-indigo-600 hover:bg-indigo-500
text-white text-xs px-3 py-1.5 rounded-lg flex items-center gap-1
transition font-medium">
              <svg class="w-3.5 h-3.5
fill-current" viewBox="0 0 24 24"><path d="M8 5v14l11-7z"/></svg>
Play Sim
            </button>
            <button onclick="resetPlayback()"
class="bg-slate-800 hover:bg-slate-700 text-slate-300 text-xs px-3
py-1.5 rounded-lg flex items-center gap-1 transition font-medium">
              <svg class="w-3.5 h-3.5"
fill="none" viewBox="0 0 24 24" stroke="currentColor"><path
stroke-linecap="round" stroke-linejoin="round" stroke-width="2"
d="M4 4v5h.582m15.356 2A8.001 8.001 0 1121.253 8H18" /></svg> Reset
            </button>
          </div>
          <div class="flex gap-1.5 justify-end">
            <button onclick="jumpToSeconds(300)"
class="text-[10px] bg-slate-900 border border-slate-800
hover:border-slate-700 text-slate-300 px-2 py-0.5 rounded
font-mono">10m Rem</button>
            <button onclick="jumpToSeconds(390)"
class="text-[10px] bg-slate-900 border border-slate-800
hover:border-slate-700 text-slate-300 px-2 py-0.5 rounded
font-mono">8.5m Rem</button>
            <button onclick="jumpToSeconds(675)"
class="text-[10px] bg-slate-900 border border-slate-800
hover:border-slate-700 text-slate-300 px-2 py-0.5 rounded
font-mono">3:45 Rem</button>
          </div>

```

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        </div>
    </div>

    <!-- NEW: OVERSIZED HIGH-VISIBILITY MOBILE DECISION
    ADVISOR HUD -->
    <div id="activeDirectiveBox" class="bg-slate-950
border-2 border-slate-800 p-5 rounded-2xl flex flex-col md:flex-row
items-center justify-between gap-4 mt-1 transition-all duration-300
shadow-xl">
        <div class="flex items-center gap-4 w-full
md:w-auto">
            <div id="directiveIconBox" class="h-14 w-14
rounded-2xl bg-slate-800 flex items-center justify-center font-mono
text-xl font-black text-slate-300 shadow-inner shrink-0">
                --
            </div>
            <div class="flex-1">
                <span class="text-[10px] font-mono
font-bold tracking-widest text-slate-400 block uppercase mb-0.5">Ⓢ
AUTOMATED STRATEGY FINAL ANSWER</span>
                <span id="directiveText" class="text-xs
font-semibold text-slate-300 leading-snug block">Initialize the
countdown sequencer to activate continuous execution
parameters.</span>
            </div>
        </div>
        <div class="w-full md:w-auto text-center
shrink-0">
            <span id="directiveStatusBadge"
class="w-full md:w-auto inline-block text-xl font-mono font-black
uppercase tracking-wider px-6 py-3 rounded-xl bg-slate-900
text-slate-500 border border-slate-800 shadow-md">
                NO DATA
            </span>
        </div>
    </div>

    <!-- Manual Time Slider Scrub Bar -->
    <div class="flex flex-col gap-1 mt-2">
        <input type="range" id="timeScrubber" min="0"
max="900" value="0" class="w-full h-1.5 bg-slate-800 rounded-lg

```

```

appearance-none cursor-pointer accent-indigo-500">
    <div class="flex justify-between text-[10px]
text-slate-500 font-mono">
        <span>15:00 START</span>
        <span>10:00 MARK</span>
        <span>08:30 MARK</span>
        <span>03:45 MARK</span>
        <span>00:00 END</span>
    </div>
</div>

    <!-- Platform Type Selection Matrix & Realtime
Window Advisor -->
    <div class="mt-1 flex flex-col gap-2">
        <div class="flex items-center justify-between
text-xs bg-slate-950 p-2.5 rounded-xl border border-slate-800/80">
            <label class="text-slate-400 font-mono
font-medium flex items-center gap-1.5">
                <svg class="w-3.5 h-3.5
text-indigo-400" fill="none" viewBox="0 0 24 24"
stroke="currentColor">

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                <path stroke-linecap="round"
stroke-linejoin="round" stroke-width="2" d="M21 12a9 9 0 01-9
9m9-9a9 9 0 00-9-9m9 9H3m9 9a9 9 0 01-9-9m9 9c1.657 0 3-4.03
3-9s-1.343-9-3-9m0 18c-1.657 0 3-4.03-3-9s1.343-9 3-9m-9 9h18" />
            </svg>
            Target Platform Deadlines:
        </label>
        <select id="platformTypeSelect"
onchange="executeCalculations(); syncTimerDisplay();
saveAppState();" class="bg-slate-900 border border-slate-700
rounded px-2.5 py-1 text-slate-200 font-mono text-[11px]
focus:outline-none focus:border-indigo-500">
            <option value="continuous"
selected>Continuous Live (Polymarket / Kalshi)</option>
            <option value="prelocked">Pre-Locked
Round (PancakeSwap Style)</option>
        </select>
    </div>

```

```

    <!-- Mini Movement Latency Predictor Tracker
-->
    <div class="grid grid-cols-1 md:grid-cols-2
gap-2">

```

```

        <div id="optimalWindowBanner" class="p-2.5
rounded-xl border text-center text-xs font-mono font-bold
transition-all duration-200 bg-slate-950 border-slate-800
text-slate-500 flex items-center justify-center">
            ANALYZING TEMPORAL RISK DATA...
        </div>
        <div id="movementLatencyPredictorBox"
class="p-2.5 rounded-xl border bg-slate-950 border-slate-800
text-slate-400 font-mono text-[11px] flex flex-col justify-center
gap-0.5">
            <div class="flex justify-between">
                <span>Action Delta Limit:</span>
                <span id="latencyBudgetVal"
class="text-emerald-400 font-bold">2.50s (SAFE)</span>

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```

            </div>
            <div class="w-full bg-slate-900 h-1
rounded-full overflow-hidden mt-1">
                <div id="latencyProgressBar"
class="h-full bg-emerald-500 transition-all duration-300"
style="width: 100%;"></div>
            </div>
        </div>
    </div>
</div>
</div>
</div>

```

```

        <!-- Infrastructure Status Flow Chart Component -->
        <div class="bg-slate-900 border border-slate-800
rounded-2xl p-5 shadow-sm flex flex-col gap-3">
            <div class="flex items-center justify-between
border-b border-slate-800 pb-2">
                <div class="flex items-center gap-2">
                    <svg class="w-5 h-5 text-indigo-400"
fill="none" viewBox="0 0 24 24" stroke="currentColor">
                        <path stroke-linecap="round"
stroke-linejoin="round" stroke-width="2" d="M19 11H5m14 0a2 2 0 012
2v6a2 2 0 01-2 2H5a2 2 0 01-2-2v-6a2 2 0 012-2m14 0V9a2 2 0
00-2-2M5 11V9a2 2 0 012-2m0 0V5a2 2 0 012-2h6a2 2 0 012 2v2M7 7h10"
/>
                    </svg>
                    <h3 class="font-semibold text-slate-200
text-sm">Live System Pipeline & Infrastructure Flow Chart</h3>
                </div>
                <button onclick="simulateNetworkOutageSpike()"

```

```
class="text-[10px] bg-slate-950 hover:bg-slate-800 border
border-slate-800 text-amber-400 px-2 py-0.5 rounded font-mono
transition">
```

```
        Trigger Outage Simulation
    </button>
</div>
```

```
        <!-- Custom Inline SVG Pipeline Graph Flow Nodes
Canvas -->
```

```
        <div class="bg-slate-950 p-4 rounded-xl border
border-slate-800/80 flex flex-col gap-3 relative overflow-hidden">
            <div class="flex justify-between items-center
text-[10px] font-mono font-bold text-slate-500 uppercase
tracking-wider mb-1">
                <span>Input Origin</span>
                <span>Analytical Core</span>
                <span>Exchange API Gateway</span>
                <span>On-Chain Settlement</span>
            </div>
```

```
        <!-- Flow Diagram Vector Path Graphic Row -->
        <div class="relative w-full h-16 flex
items-center justify-between z-10 px-2">
```

```
            <!-- Node 1: Mobile UI Focus Stance State
-->
```

```
                <div id="flowNode-ui" class="h-10 w-20
rounded-lg border border-emerald-500/40 bg-emerald-500/10
text-emerald-400 font-mono text-[10px] flex flex-col items-center
justify-center text-center transition-all duration-300 font-bold
shadow-sm">
```

```
                    <span>MOBILE UI</span>
                    <span class="text-[8px] opacity-80"
id="flowStatus-ui">ACTIVE</span>
```

```
        </div>
```

```
        <!-- Connector 1 -->
        <div class="flex-1 h-1 mx-1 relative">
            <svg class="w-full h-4
overflow-visible" preserveAspectRatio="none">
```

```
                <line x1="0%" y1="2" x2="100%"
```

```
y2="2" id="flowLine-1" class="stroke-emerald-500 flow-line-active"
stroke-width="2"/>
```

```
</svg>
</div>
```

```
<!-- Node 2: Core Matrix Probability
Calculation Processor -->
```

```
<div id="flowNode-core" class="h-10 w-20
rounded-lg border border-emerald-500/40 bg-emerald-500/10
text-emerald-400 font-mono text-[10px] flex flex-col items-center
justify-center text-center transition-all duration-300 font-bold">
```

```
<span>CORE ENG</span>
<span class="text-[8px] opacity-80"
id="flowStatus-core">COMPUTING</span>
</div>
```

```
<!-- Connector 2 -->
<div class="flex-1 h-1 mx-1 relative">
<svg class="w-full h-4
overflow-visible" preserveAspectRatio="none">
<line x1="0%" y1="2" x2="100%"
y2="2" id="flowLine-2" class="stroke-emerald-500 flow-line-active"
stroke-width="2"/>
```

```
</svg>
</div>
```

```
<!-- Node 3: Coinbase Data Sync Feed &
Exchange Gateway Status -->
```

```
<div id="flowNode-coinbase" class="h-10
w-20 rounded-lg border border-emerald-500/40 bg-emerald-500/10
text-emerald-400 font-mono text-[10px] flex flex-col items-center
justify-center text-center transition-all duration-300 font-bold">
```

```
<span>COINBASE</span>
<span class="text-[8px] opacity-80"
id="flowStatus-coinbase">CONNECTED</span>
</div>
```

```
<!-- Connector 3 -->
<div class="flex-1 h-1 mx-1 relative">
<svg class="w-full h-4
overflow-visible" preserveAspectRatio="none">
<line x1="0%" y1="2" x2="100%"
y2="2" id="flowLine-3" class="stroke-emerald-500 flow-line-active"
stroke-width="2"/>
```

```
        </svg>
    </div>
```

```
        <!-- Node 4: Contract Router Smart Contract
Settlement Node -->
```

```
        <div id="flowNode-contract" class="h-10
w-20 rounded-lg border border-emerald-500/40 bg-emerald-500/10
text-emerald-400 font-mono text-[10px] flex flex-col items-center
justify-center text-center transition-all duration-300 font-bold">
            <span>ROUTER</span>
```

```
            <span class="text-[8px] opacity-80"
id="flowStatus-contract">READY</span>
        </div>
    </div>
```

```
        <!-- Flow Alert Summary HUD Banner -->
        <div id="flowAlertHudBanner" class="mt-1
text-center py-1.5 px-3 rounded-lg text-xs font-mono font-medium
bg-slate-900 border border-slate-800 text-slate-400">
            ⚡ Infrastructure diagnostics reporting
nominal 100% telemetry connection uptime.
        </div>
    </div>
</div>
```

```
        <!-- Mathematical Probability Output Meter Visualizer
-->
        <div class="bg-slate-900 border border-slate-800
rounded-2xl p-5 shadow-sm flex-1 flex flex-col gap-5
justify-between">
            <div class="flex items-center justify-between
border-b border-slate-800 pb-3">
                <div class="flex items-center gap-2">
                    <svg class="w-5 h-5 text-indigo-400"
fill="none" viewBox="0 0 24 24" stroke="currentColor">
                        <path stroke-linecap="round"
stroke-linejoin="round" stroke-width="2" d="M9 19v-6a2 2 0
00-2-2H5a2 2 0 00-2 2v6a2 2 0 002 2h2a2 2 0 002 2zm0 0V9a2 2 0
012-2h2a2 2 0 012 2v10m-6 0a2 2 0 002 2h2a2 2 0 002 2m0 0V5a2 2 0
012-2h2a2 2 0 012 2v14m-6 0a2 2 0 002 2h2a2 2 0 002 2" />
                    </svg>
                    <h3 class="font-semibold
text-slate-200">Probability Computations (Rolling)</h3>
```

```

        </div>
        <span class="text-xs font-mono
text-slate-500">Updated Dynamically</span>
    </div>

```

```

        <!-- Main Dual Probability Meters -->
        <div class="grid grid-cols-2 gap-4 items-center">
            <!-- OVER Meter -->
            <div class="bg-slate-950 p-4 rounded-xl border
border-slate-800 flex flex-col items-center relative
overflow-hidden">
                <div class="absolute top-0 left-0 h-1
bg-emerald-500 transition-all duration-300" id="overProbBarTop"
style="width: 50%;"></div>
                <span class="text-xs font-mono
text-emerald-400 font-semibold mb-1">OVER PROBABILITY</span>
                <span id="overProbabilityDisplay"
class="text-4xl font-bold font-mono text-white
tracking-tight">50.0%</span>
                <p class="text-[10px] text-slate-500 mt-2
text-center" id="overProbBreakdown">Base: 50.0% | Momentum: +0%</p>
            </div>

```

```

            <!-- UNDER Meter -->
            <div class="bg-slate-950 p-4 rounded-xl border
border-slate-800 flex flex-col items-center relative
overflow-hidden">
                <div class="absolute top-0 left-0 h-1
bg-rose-500 transition-all duration-300" id="underProbBarTop"
style="width: 50%;"></div>
                <span class="text-xs font-mono
text-rose-400 font-semibold mb-1">UNDER PROBABILITY</span>
                <span id="underProbabilityDisplay"
class="text-4xl font-bold font-mono text-white
tracking-tight">50.0%</span>

```

```

                <p class="text-[10px] text-slate-500 mt-2
text-center" id="underProbBreakdown">Base: 50.0% | Momentum:
+0%</p>
            </div>
        </div>

```

```

        <!-- Probability Balance Sliding visual meter -->
        <div class="w-full bg-slate-950 h-4 rounded-full
border border-slate-800 overflow-hidden relative mt-2 flex">
            <div id="overBarSlider" class="h-full
bg-gradient-to-r from-emerald-600 to-emerald-400 transition-all
duration-300" style="width: 50%;"></div>
            <div id="underBarSlider" class="h-full
bg-gradient-to-r from-rose-400 to-rose-600 transition-all
duration-300" style="width: 50%;"></div>
            <div class="absolute inset-0 flex items-center
justify-between px-3 text-[9px] font-mono font-bold text-slate-200
pointer-events-none mix-blend-difference">
                <span>OVER bias</span>
                <span>UNDER bias</span>
            </div>
        </div>
    </div>
</section>

```

```

    <!-- Right Column: Specific Milestone Matrix & Action Log
(3 Columns Wide) -->
    <section class="xl:col-span-3 flex flex-col gap-6">

```

```

        <!-- System Performance Ledger Card -->
        <div class="bg-slate-900 border border-slate-800
rounded-2xl p-5 shadow-sm flex flex-col gap-4">
            <div class="flex items-center justify-between
border-b border-slate-800 pb-1">
                <div class="flex items-center gap-2">
                    <svg class="w-5 h-5 text-indigo-400"
fill="none" viewBox="0 0 24 24" stroke="currentColor">
                        <path stroke-linecap="round"
stroke-linejoin="round" stroke-width="2" d="M16
8v8m-4-5v5m-4-2v2m-2 4h12a2 2 0 02-2V6a2 2 0 00-2-2H6a2 2 0 00-2
2v12a2 2 0 002 2z" />
                    </svg>
                    <h3 class="font-semibold
text-slate-200">Performance Ledger</h3>
                </div>
                <span id="winRateBadge" class="text-xs
font-mono font-bold px-2 py-0.5 rounded bg-indigo-500/10
text-indigo-400 border border-indigo-500/20">WR: 0.0%</span>
            </div>

```

```

        <!-- Win/Loss Statistics Counters Grid -->
        <div class="grid grid-cols-3 gap-2 text-center">
            <div class="bg-slate-950 p-2 rounded-xl border
border-slate-800/60">
                <div class="text-[9px] uppercase font-mono
text-slate-500">Wins</div>
                <div id="ledgerWins" class="text-base
font-bold font-mono text-emerald-400 mt-0.5">0</div>
            </div>
            <div class="bg-slate-950 p-2 rounded-xl border
border-slate-800/60">

```

```

                <div class="text-[9px] uppercase font-mono
text-slate-500">Losses</div>
                <div id="ledgerLosses" class="text-base
font-bold font-mono text-rose-400 mt-0.5">0</div>
            </div>
            <div class="bg-slate-950 p-2 rounded-xl border
border-slate-800/60">
                <div class="text-[9px] uppercase font-mono
text-slate-500">Total</div>
                <div id="ledgerTotal" class="text-base
font-bold font-mono text-slate-300 mt-0.5">0</div>
            </div>
        </div>

```

```

        <!-- Transmission Health Tracker Section -->
        <div class="border-t border-slate-800/80 pt-3 flex
flex-col gap-2">
            <div class="flex justify-between items-center">
                <span class="text-xs font-semibold
text-slate-300 flex items-center gap-1">
                     Transmission Health
                </span>
                <span id="txRateBadge" class="text-[10px]
font-mono font-bold px-1.5 py-0.5 rounded bg-amber-500/10
text-amber-400 border border-amber-500/20">TX RATE: 0.0%</span>
            </div>
            <div class="grid grid-cols-3 gap-2
text-center">
                <div class="bg-slate-950 p-1.5 rounded-xl
border border-slate-800/40">

```

```

        <div class="text-[8px] uppercase
font-mono text-slate-500">Confirmed</div>
        <div id="txSuccess" class="text-xs
font-bold font-mono text-emerald-400 mt-0.5">0</div>
    </div>
    <div class="bg-slate-950 p-1.5 rounded-xl
border border-slate-800/40">
        <div class="text-[8px] uppercase
font-mono text-slate-500">Slipped</div>
        <div id="txFailed" class="text-xs
font-bold font-mono text-rose-400 mt-0.5">0</div>
    </div>
    <div class="bg-slate-950 p-1.5 rounded-xl
border border-slate-800/40">
        <div class="text-[8px] uppercase
font-mono text-slate-500">Total</div>
        <div id="txTotal" class="text-xs
font-bold font-mono text-slate-300 mt-0.5">0</div>
    </div>
</div>
</div>

```

```

        <!-- Manual Resolution Workspace Section -->
        <div class="bg-slate-950 p-3 rounded-xl border
border-slate-800/80 flex flex-col gap-2 border-t
border-slate-800/40 pt-3">
            <span class="text-[11px] text-slate-400
font-medium">Settle Current Active Cycle:</span>
            <div class="grid grid-cols-2 gap-2">
                <button
onclick="settleCurrentCycle('OVER')" class="bg-emerald-600/25
hover:bg-emerald-600/40 text-emerald-400 text-xs py-1.5 rounded-lg
border border-emerald-500/20 transition font-mono font-bold
tracking-wide">

```

```

                SETTLE OVER
            </button>
            <button
onclick="settleCurrentCycle('UNDER')" class="bg-rose-600/25
hover:bg-rose-600/40 text-rose-400 text-xs py-1.5 rounded-lg border
border-rose-500/20 transition font-mono font-bold tracking-wide">
                SETTLE UNDER

```

```

        </button>
      </div>
    </div>
    <!-- Local Cache Clear Operations Pad -->
    <button onclick="purgeLocalCacheSession()"
class="w-full bg-slate-950 hover:bg-slate-900 border
border-slate-800 text-slate-400 text-[10px] font-mono py-1
rounded-lg transition">
      🗑️ Clear Saved Local Cache
    </button>
  </div>

```

```

    <!-- Target Milestones Tracker Panel -->
    <div class="bg-slate-900 border border-slate-800
rounded-2xl p-5 shadow-sm flex flex-col gap-4">
      <div class="flex items-center gap-2 border-b
border-slate-800 pb-3">
        <svg class="w-5 h-5 text-indigo-400"
fill="none" viewBox="0 0 24 24" stroke="currentColor">
          <path stroke-linecap="round"
stroke-linejoin="round" stroke-width="2" d="M9 5H7a2 2 0 0-2
2v12a2 2 0 02 2h10a2 2 0 02-2V7a2 2 0 0-2-2h-2M9 5a2 2 0 02
2h2a2 2 0 02-2M9 5a2 2 0 012-2h2a2 2 0 012 2" />
        </svg>

```

```

      <h3 class="font-semibold
text-slate-200">Milestone Signal Registry</h3>
    </div>

```

```

    <div class="flex flex-col gap-3">
      <div id="msCard-10_min" class="bg-slate-950
border border-slate-800 rounded-xl p-3 flex flex-col gap-2
transition duration-200">
        <div class="flex justify-between
items-center">
          <span class="text-xs font-bold
font-mono text-slate-300">10:00 Remaining</span>
          <span id="msBadge-10_min"
class="text-[10px] font-mono bg-slate-900 text-slate-500 border
border-slate-800 px-1.5 py-0.5 rounded font-bold
uppercase">PENDING</span>
        </div>
        <div class="flex justify-between
items-center text-[11px] font-mono text-slate-400">
          <span>Stance Calc: <span>

```

```
id="msStance-10_min" class="text-slate-300">--</span></span>
      <span>Prob: <span id="msProb-10_min"
class="text-slate-300">--</span></span>
    </div>
  </div>
```

```
    <div id="msCard-8_5_min" class="bg-slate-950
border border-slate-800 rounded-xl p-3 flex flex-col gap-2
transition duration-200">
      <div class="flex justify-between
items-center">
        <span class="text-xs font-bold
font-mono text-slate-300">08:30 Remaining</span>
        <span id="msBadge-8_5_min"
class="text-[10px] font-mono bg-slate-900 text-slate-500 border
border-slate-800 px-1.5 py-0.5 rounded font-bold
uppercase">PENDING</span>
```

```
      </div>
      <div class="flex justify-between
items-center text-[11px] font-mono text-slate-400">
        <span>Stance Calc: <span
id="msStance-8_5_min" class="text-slate-300">--</span></span>
        <span>Prob: <span id="msProb-8_5_min"
class="text-slate-300">--</span></span>
      </div>
    </div>
```

```
    <div id="msCard-3_75_min" class="bg-slate-950
border border-slate-800 rounded-xl p-3 flex flex-col gap-2
transition duration-200">
      <div class="flex justify-between
items-center">
        <span class="text-xs font-bold
font-mono text-slate-300">03:45 Remaining</span>
        <span id="msBadge-3_75_min"
class="text-[10px] font-mono bg-slate-900 text-slate-500 border
border-slate-800 px-1.5 py-0.5 rounded font-bold
uppercase">PENDING</span>
      </div>
      <div class="flex justify-between
items-center text-[11px] font-mono text-slate-400">
        <span>Stance Calc: <span
id="msStance-3_75_min" class="text-slate-300">--</span></span>
        <span>Prob: <span id="msProb-3_75_min"
```

```

class="text-slate-300">--</span></span>
      </div>
    </div>
  </div>
</div>

```

```

      <!-- Event Log Box Component -->
      <div class="bg-slate-900 border border-slate-800
rounded-2xl p-5 shadow-sm flex-1 flex flex-col gap-3
min-h-[180px]">
        <div class="flex items-center justify-between
border-b border-slate-800 pb-2">
          <span class="text-xs font-bold uppercase
tracking-wider text-slate-400 font-mono">System Events Log</span>
          <button onclick="clearLogs()"
class="text-[10px] text-slate-500 hover:text-slate-300
font-mono">Clear</button>
        </div>
        <div id="logConsole" class="flex-1 overflow-y-auto
bg-slate-950 rounded-xl border border-slate-800 p-3 font-mono
text-[11px] text-slate-400 flex flex-col gap-1.5 leading-relaxed
max-h-[100px]">
          <div class="text-slate-500">[00:00:00]
Probability environment initialized.</div>
        </div>
      </div>
    </section>
  </main>

```

```

      <!-- TAB CONTAINER TWO: ADVANCED BATCH BACKTESTING WORKSPACE
MATRIX -->

```

```

      <main id="backtestWorkspaceView" class="flex-1 max-w-[1600px]
w-full mx-auto p-4 lg:p-6 grid grid-cols-1 lg:grid-cols-12 gap-6
overflow-y-auto hidden">

```

```

      <!-- Left Side Backtest Strategy Controllers Setup -->
      <section class="lg:col-span-4 flex flex-col gap-6">
        <div class="bg-slate-900 border border-slate-800
rounded-2xl p-5 flex flex-col gap-4">

```

```

      <div class="flex items-center gap-2 border-b

```

```

border-slate-800 pb-3">
    <svg class="w-5 h-5 text-indigo-400"
fill="none" viewBox="0 0 24 24" stroke="currentColor">
        <path stroke-linecap="round"
stroke-linejoin="round" stroke-width="2" d="M19.428 15.428a2 2 0
00-1.022-.547l-2.387-.477a6 6 0 00-3.86.517l-.318.158a6 6 0
01-3.86.517L6.05 15.21a2 2 0 00-1.806.547M8 4h8l-1 1v5.172a2 2 0
00.586 1.414l5 5c1.26 1.26.367 3.414-1.415 3.414H4.828c-1.782
0-2.674-2.154-1.414-3.414l5-5A2 2 0 009 10.172V5L8 4z" />
    </svg>
    <h2 class="font-semibold
text-slate-200">Backtest Parameter Matrix</h2>
</div>

<!-- Sample Size Selector -->
<div class="flex flex-col gap-1.5">
    <label class="text-xs text-slate-300
font-medium">Batch Interval Samples Length</label>
    <select id="btSampleSize" class="w-full
bg-slate-950 border border-slate-800 rounded-xl px-3 py-2 text-sm
text-slate-200 font-mono focus:outline-none
focus:border-indigo-500">
        <option value="50">50 Sequential
Blocks</option>
        <option value="100" selected>100 Sequential
Blocks</option>
        <option value="250">250 Sequential
Blocks</option>
        <option value="500">500 Sequential
Blocks</option>
    </select>
</div>

<!-- Market Bias Conditioner -->

<div class="flex flex-col gap-1.5">
    <label class="text-xs text-slate-300
font-medium">Historical Market Environment Bias</label>
    <select id="btMarketBias" class="w-full
bg-slate-950 border border-slate-800 rounded-xl px-3 py-2 text-sm
text-slate-200 font-mono focus:outline-none
focus:border-indigo-500">
        <option value="bullish">High Bullish
Volatility (65% Over Bias)</option>
        <option value="bearish">Aggressive Selloffs

```

```

(65% Under Bias)</option>
      <option value="ranging" selected>Mean
Reverting Ranging / Chop (Balanced 50/50)</option>
    </select>
  </div>

  <!-- Core Strategy Inputs Summary Replication -->
  <div class="bg-slate-950 p-4 rounded-xl border
border-slate-800/80 flex flex-col gap-3 text-xs">
    <span class="font-medium text-slate-400 block
uppercase tracking-wider font-mono text-[10px]">Active Rule Sets
Applied</span>
    <div class="flex justify-between">
      <span class="text-slate-500">Confidence
Barrier:</span>
      <span id="btRefThreshold" class="font-mono
text-indigo-400 font-bold">55%</span>
    </div>
    <div class="flex justify-between">
      <span class="text-slate-500">Intra-Candle
Momentum Factor:</span>
      <span id="btRefMomentum" class="font-mono
text-emerald-400 font-bold">+20%</span>
    </div>
    <div class="flex justify-between">
      <span class="text-slate-500">Decay Weight
Matrix Factor:</span>
      <span class="font-mono
text-slate-300">Linear Decay (+15%/step)</span>
    </div>
  </div>

  <button onclick="executeBulkBacktest()"
class="w-full bg-indigo-600 hover:bg-indigo-500 text-white
font-semibold py-3 px-4 rounded-xl transition duration-150 flex
items-center justify-center gap-2 shadow-lg shadow-indigo-600/10">
    <svg class="w-4 h-4" fill="none" viewBox="0 0
24 24" stroke="currentColor">
      <path stroke-linecap="round"
stroke-linejoin="round" stroke-width="2" d="M14.752
11.168l-3.197-2.132A1 1 0 010 9.87v4.263a1 1 0
001.555.832l3.197-2.132a1 1 0 00-1.664z" />
      <path stroke-linecap="round"
stroke-linejoin="round" stroke-width="2" d="M21 12a9 9 0 11-18 0 9

```

```

9 0 0118 0z" />
        </svg>
        Run Matrix Simulation
    </button>
</div>
</section>

<!-- Right Side Comprehensive Metrics Panel Dash -->
<section class="lg:col-span-8 flex flex-col gap-6">

    <!-- Global Backtest Performance Stats Dashboard Rows
-->
    <div class="bg-slate-900 border border-slate-800
rounded-2xl p-5 flex flex-col gap-4">
        <div class="flex justify-between items-center
border-b border-slate-800 pb-3">
            <h3 class="font-semibold
text-slate-200">Aggregate Backtest Metrics</h3>
            <span id="btSummaryStatus" class="text-[10px]
font-mono font-bold uppercase tracking-wider px-2 py-0.5
bg-slate-950 border border-slate-800 text-slate-500 rounded">No
Data Loaded</span>
        </div>

        <div class="grid grid-cols-2 md:grid-cols-4 gap-4">
            <div class="bg-slate-950 p-4 rounded-xl border
border-slate-800/80">
                <span class="text-[10px] text-slate-500
font-mono uppercase block">Total Cycles Ran</span>
                <div id="btMetricTotal" class="text-2xl
font-bold font-mono text-slate-200 mt-1">0</div>
            </div>
            <div class="bg-slate-950 p-4 rounded-xl border
border-slate-800/80">
                <span class="text-[10px] text-slate-500
font-mono uppercase block">Strategy Win Rate</span>
                <div id="btMetricWinRate" class="text-2xl
font-bold font-mono text-slate-400 mt-1">0.0%</div>
            </div>
            <div class="bg-slate-950 p-4 rounded-xl border
border-slate-800/80">
                <span class="text-[10px] text-slate-500
font-mono uppercase block">Total Wins</span>

```

```

        <div id="btMetricWins" class="text-2xl
font-bold font-mono text-emerald-400 mt-1">0</div>
    </div>
    <div class="bg-slate-950 p-4 rounded-xl border
border-slate-800/80">
        <span class="text-[10px] text-slate-500
font-mono uppercase block">Total Losses</span>
        <div id="btMetricLosses" class="text-2xl
font-bold font-mono text-rose-400 mt-1">0</div>
    </div>
</div>

```

```

    <!-- Milestone Specific Hit Rates breakdown grids
-->
    <div class="grid grid-cols-1 md:grid-cols-3 gap-3
mt-1">
        <div class="p-3 bg-slate-950 rounded-xl border
border-slate-800/60 flex items-center justify-between">
            <span class="text-xs font-mono
text-slate-400">10m Mark Hit-Rate:</span>
            <span id="btHit-10m" class="font-bold
font-mono text-slate-300">--</span>
        </div>
        <div class="p-3 bg-slate-950 rounded-xl border
border-slate-800/60 flex items-center justify-between">
            <span class="text-xs font-mono
text-slate-400">8.5m Mark Hit-Rate:</span>
            <span id="btHit-8_5m" class="font-bold
font-mono text-slate-300">--</span>
        </div>
        <div class="p-3 bg-slate-950 rounded-xl border
border-slate-800/60 flex items-center justify-between">

```

```

            <span class="text-xs font-mono
text-slate-400">3:45m Mark Hit-Rate:</span>
            <span id="btHit-3_75m" class="font-bold
font-mono text-slate-300">--</span>
        </div>
    </div>
</div>

```

```

    <!-- Individual Cycle Output Log Terminal Workspace

```

Registry Lists -->

```
<div class="bg-slate-900 border border-slate-800
rounded-2xl p-5 flex-1 flex-col gap-3 min-h-[350px]">
  <span class="text-xs font-bold text-slate-400
uppercase font-mono tracking-wider border-b border-slate-800 pb-2
block">
```

Matrix Item Registry Output Feed

```
</span>
```

```
<div class="flex-1 overflow-y-auto max-h-[400px]
border border-slate-800 rounded-xl bg-slate-950">
```

```
<table class="w-full text-left font-mono
text-xs border-collapse">
```

```
<thead class="bg-slate-900 text-slate-400
sticky top-0 border-b border-slate-800">
```

```
<tr>
```

```
<th class="p-3">Block ID</th>
```

```
<th class="p-3">Historical Inputs
```

```
Sequence</th>
```

```
<th class="p-3 text-center">10m
```

```
Sig</th>
```

```
<th class="p-3 text-center">8.5m
```

```
Sig</th>
```

```
<th class="p-3 text-center">3.45m
```

```
Sig</th>
```

```
<th class="p-3">Actual</th>
```

```
<th class="p-3 text-right">Block
```

```
Performance</th>
```

```
</tr>
```

```
</thead>
```

```
<tbody id="btRegistryTableBody"
class="text-slate-300 divide-y divide-slate-900">
```

```
<tr>
```

```
<td colspan="7" class="text-center
p-8 text-slate-500">Configure parameters matrices on left side and
click compute execution to process logs.</td>
```

```
</tr>
```

```
</tbody>
```

```
</table>
```

```
</div>
```

```
</div>
```

```
</section>
```

```
</main>
```

```
<!-- Footer Status Information -->
<footer class="bg-slate-900/40 border-t border-slate-800 px-6
py-3 text-center text-xs text-slate-500">
```

```
Continuous Mathematical Analytics Prototype • Automated
Historical Matrix Simulation Active
</footer>
```

```
<!-- Core Analytical Logic & State Controller Script -->
<script>
  // State variables initialization
  let historicalResults = [1, 1, 0, 1, 0, 1];
  let userStance = 'OVER';
  let confidenceThreshold = 0.55;
  let microMomentum = 0.20;
  let elapsedSeconds = 0;
  let isSimPlaying = false;
  let simIntervalId = null;
  let timerMode = 'sim';
  let activeBidValue = 0.75;
```

```
  // User submission lag profiling data parameters
  let userExecutionLag = 1.50;
```

```
  // Ledger metrics state initialization
  let totalWins = 0;
  let totalLosses = 0;
```

```
  // Transmission Ledger Variables
  let totalTx = 0;
  let successfulTx = 0;
  let failedTx = 0;
```

```
  // Circuit Breaker Control Variable
  let isCoinbaseDown = false;
```

```
  // Milestone configuration definitions
  const MILESTONES = {
    "10_min": 300,
    "8_5_min": 390,
    "3_75_min": 675
```

```
};
```

```
let lockedMilestoneData = {  
  "10_min": null,  
  "8_5_min": null,  
  "3_75_min": null  
};
```

```
window.addEventListener('DOMContentLoaded', () => {  
  loadAppStateFromLocalStorage(); // Pull cached values  
before initializing DOM configurations  
  renderHistoricalBlocks();  
  initControlEventsListeners();  
  executeCalculations();  
  startSystemClock();  
  startTickerTapeStream();  
  initVisibilityFocusHandler();  
});
```

```
// LOCAL STORAGE STATE WRITER COMPONENT  
function saveAppState() {  
  try {
```

```
    const cachePayload = {  
      historicalResults,  
      userStance,  
      confidenceThreshold,  
      microMomentum,  
      activeBidValue,  
      totalWins,  
      totalLosses,  
      totalTx,  
      successfulTx,  
      failedTx,  
      platformType:  
document.getElementById('platformTypeSelect').value  
    };  
    localStorage.setItem('prob_engine_session_state',  
JSON.stringify(cachePayload));  
  } catch (err) {  
    console.error("Failed to sync structural
```

```
configuration to local caching directory:", err);
    }
}
```

```
// LOCAL STORAGE STATE READER COMPONENT
function loadAppStateFromLocalStorage() {
    try {
        const cachedData =
localStorage.getItem('prob_engine_session_state');
        if (cachedData) {
            const parsed = JSON.parse(cachedData);

            if (Array.isArray(parsed.historicalResults))
historicalResults = parsed.historicalResults;
            if (parsed.userStance) userStance =
parsed.userStance;
            if (typeof parsed.confidenceThreshold ===
'number') confidenceThreshold = parsed.confidenceThreshold;
            if (typeof parsed.microMomentum === 'number')
microMomentum = parsed.microMomentum;
            if (typeof parsed.activeBidValue === 'number')
activeBidValue = parsed.activeBidValue;
            if (typeof parsed.totalWins === 'number')
totalWins = parsed.totalWins;
            if (typeof parsed.totalLosses === 'number')
totalLosses = parsed.totalLosses;
            if (typeof parsed.totalTx === 'number') totalTx
= parsed.totalTx;
            if (typeof parsed.successfulTx === 'number')
successfulTx = parsed.successfulTx;
            if (typeof parsed.failedTx === 'number')
failedTx = parsed.failedTx;

            // Restore control slider input position
            configurations programmatically
```

```
document.getElementById('thresholdSlider').value =
Math.round(confidenceThreshold * 100);
```

```
document.getElementById('thresholdValueDisplay').innerText =
```

```

Math.round(confidenceThreshold * 100) + '%';

        document.getElementById('momentumSlider').value
= Math.round(microMomentum * 100);
        const prefix = microMomentum >= 0 ? '+' : '';

document.getElementById('momentumValueDisplay').innerText = prefix
+ Math.round(microMomentum * 100) + '%';

        document.getElementById('manualBidInput').value
= activeBidValue.toFixed(2);

document.getElementById('activeBidDisplay').innerText =
activeBidValue.toFixed(2) + ' ETH';

document.getElementById('transmitBidActionBtn').innerText = `🚀
TRANSMIT BID NOW (${activeBidValue.toFixed(2)} ETH)`;

        if (parsed.platformType)
document.getElementById('platformTypeSelect').value =
parsed.platformType;

        // Rehydrate performance tracking metric tables
rows data
        document.getElementById('ledgerWins').innerText
= totalWins;

document.getElementById('ledgerLosses').innerText = totalLosses;
        const combinedLedger = totalWins + totalLosses;

document.getElementById('ledgerTotal').innerText = combinedLedger;
        if (combinedLedger > 0) {

document.getElementById('winRateBadge').innerText = `WR:
${(totalWins / combinedLedger * 100).toFixed(1)}%`;
        }

        updateTxLedgerDisplay();
        syncStanceUI();
        addLogEntry("Persistent local cache session
loaded and verified successfully.");
    }
    } catch (err) {
        console.error("Local storage lookup directory

```

```

failure:", err);
    }
}

// PURGE LOCAL STORAGE DATA CACHE
function purgeLocalCacheSession() {
    localStorage.removeItem('prob_engine_session_state');
    addLogEntry("Local memory structures purged. Reloading
defaults on next mount cycle.");
    window.location.reload();
}

// HIGH RESPONSE SCREEN-SWITCH FREEZE HANDLER (BACKGROUND
TELEMETRY MAPPING)
function initVisibilityFocusHandler() {
    document.addEventListener('visibilitychange', () => {
        const uiNode =
document.getElementById('flowNode-ui');
        const uiLabel =
document.getElementById('flowStatus-ui');
        const hud =
document.getElementById('flowAlertHudBanner');

        if (document.hidden) {
            uiNode.className = "h-10 w-20 rounded-lg border
border-amber-500/40 bg-amber-500/10 text-amber-400 font-mono
text-[10px] flex flex-col items-center justify-center text-center
transition-all duration-300 font-bold animate-pulse";
            uiLabel.innerText = "IDLE/THROT";
        } else {
            uiNode.className = "h-10 w-20 rounded-lg border
border-indigo-500/50 bg-indigo-500/20 text-indigo-400 font-mono
text-[10px] flex flex-col items-center justify-center text-center
transition-all duration-300 font-bold";
            uiLabel.innerText = "RECOVERING";

            if (!isCoinbaseDown) {
                hud.className = "mt-1 text-center py-1.5
px-3 rounded-lg text-xs font-mono font-medium bg-indigo-950 border
border-indigo-800 text-indigo-300 animate-pulse";
                hud.innerText = "⚠ Mobile app idling
detected. Resynchronizing on-chain router pipeline latency
metrics...";
            }
        }
    });
}

```

```
        addLogEntry("[INFRASTRUCTURE] Mobile viewport  
changed screens. Resyncing system core clock counters.");
```

```
        setTimeout(() => {  
            uiNode.className = "h-10 w-20 rounded-lg  
border border-emerald-500/40 bg-emerald-500/10 text-emerald-400  
font-mono text-[10px] flex flex-col items-center justify-center  
text-center transition-all duration-300 font-bold";  
            uiLabel.innerText = "ACTIVE";  
            if (!isCoinbaseDown) {  
                hud.className = "mt-1 text-center  
py-1.5 px-3 rounded-lg text-xs font-mono font-medium bg-slate-900  
border border-slate-800 text-slate-400";  
                hud.innerText = "⚡ Infrastructure  
diagnostics reporting nominal 100% telemetry connection uptime.";  
            }  
        }, 1500);  
    }  
});  
}
```

```
// COINBASE DOWNTIME & AUTOMATED CIRCUIT BREAKER TRIGGER  
function simulateNetworkOutageSpike() {  
    const cbNode =  
document.getElementById('flowNode-coinbase');  
    const cbLabel =  
document.getElementById('flowStatus-coinbase');  
    const txBtn =  
document.getElementById('transmitBidActionBtn');
```

```
    const line2 = document.getElementById('flowLine-2');  
    const line3 = document.getElementById('flowLine-3');  
    const hud =  
document.getElementById('flowAlertHudBanner');
```

```
    isCoinbaseDown = true;  
    addLogEntry("[CRITICAL] Automated Circuit Breaker  
Engaged: Coinbase API downtime detected.");
```

```
    cbNode.className = "h-10 w-20 rounded-lg border
```

```
border-rose-500/60 bg-rose-500/20 text-rose-400 font-mono
text-[10px] flex flex-col items-center justify-center text-center
transition-all duration-300 font-bold animate-bounce";
    cbLabel.innerText = "DOWN / LAG";
```

```
        txBtn.disabled = true;
        txBtn.className = "w-full mt-1.5 bg-slate-800
text-slate-500 border border-slate-700 font-extrabold py-2.5 px-4
rounded-xl cursor-not-allowed transition duration-150 flex
items-center justify-center gap-2 tracking-wide shadow-none";
        txBtn.innerText = "🔒 TRANSMISSION BLOCKED (EXCHANGE
DOWN)";
```

```
        line2.setAttribute('class', 'stroke-rose-500
flow-line-degraded');
        line3.setAttribute('class', 'stroke-rose-500
flow-line-degraded');
```

```
        hud.className = "mt-1 text-center py-1.5 px-3
rounded-lg text-xs font-mono font-bold bg-rose-950 border
border-rose-800 text-rose-400 animate-pulse";
        hud.innerText = "❌ EXTREME LAG: Coinbase connection
dropped. Transmit functionality locked to protect margin capital.";
```

```
        setTimeout(() => {
            isCoinbaseDown = false;
```

```
            cbNode.className = "h-10 w-20 rounded-lg border
border-emerald-500/40 bg-emerald-500/10 text-emerald-400 font-mono
text-[10px] flex flex-col items-center justify-center text-center
transition-all duration-300 font-bold";
            cbLabel.innerText = "CONNECTED";
```

```
            txBtn.disabled = false;
            txBtn.className = "w-full mt-1.5 bg-gradient-to-r
from-amber-500 to-orange-600 hover:from-amber-600
hover:to-orange-700 text-slate-950 font-extrabold py-2.5 px-4
rounded-xl transition duration-150 flex items-center justify-center
gap-2 tracking-wide shadow-md shadow-orange-500/10
active:scale-[0.98]";
```

```
            txBtn.innerText = `🚀 TRANSMIT BID NOW
(${activeBidValue.toFixed(2)} ETH)`;
```

```
        line2.setAttribute('class', 'stroke-emerald-500
```

```

flow-line-active');
        line3.setAttribute('class', 'stroke-emerald-500
flow-line-active');

        hud.className = "mt-1 text-center py-1.5 px-3
rounded-lg text-xs font-mono font-medium bg-slate-900 border
border-slate-800 text-slate-400";
        hud.innerText = "⚡ Infrastructure diagnostics
reporting nominal 100% telemetry connection uptime.";
        addLogEntry("[INFRASTRUCTURE] Circuit breaker
cleared. Coinbase heartbeat stabilized successfully.");
    }, 6000);
}

// Switch View Tab Controller Modality Handler
function switchViewMode(targetMode) {
    const liveTab = document.getElementById('tabLiveBtn');
    const btTab =
document.getElementById('tabBacktestBtn');
    const liveView =
document.getElementById('liveWorkspaceView');
    const btView =
document.getElementById('backtestWorkspaceView');

    if (targetMode === 'live') {
        liveTab.className = "px-4 py-1.5 text-xs
font-semibold rounded-lg bg-indigo-600 text-white transition
duration-150";
        btTab.className = "px-4 py-1.5 text-xs
font-semibold rounded-lg text-slate-400 hover:text-slate-200
transition duration-150";
        liveView.classList.remove('hidden');
        btView.classList.add('hidden');
    } else {
        btTab.className = "px-4 py-1.5 text-xs
font-semibold rounded-lg bg-indigo-600 text-white transition
duration-150";
        liveTab.className = "px-4 py-1.5 text-xs
font-semibold rounded-lg text-slate-400 hover:text-slate-200
transition duration-150";
        btView.classList.remove('hidden');
        liveView.classList.add('hidden');
    }

    document.getElementById('btRefThreshold').innerText

```

```
= (confidenceThreshold * 100).toFixed(0) + '%';
```

```
        const prefix = microMomentum >= 0 ? '+' : '';  
        document.getElementById('btRefMomentum').innerText  
= prefix + (microMomentum * 100).toFixed(0) + '%';  
    }  
}
```

```
// QUICK BID SIZE SWITCH CONTROLLER  
function setQuickBid(value) {  
    if (isCoinbaseDown) {  
        addLogEntry("[REJECTED] Cannot adjust position  
size. Execution pad is locked while exchange API is offline.");  
        return;  
    }  
    activeBidValue = value;  
    document.getElementById('activeBidDisplay').innerText =  
value.toFixed(2) + ' ETH';  
    document.getElementById('manualBidInput').value =  
value.toFixed(2);
```

```
document.getElementById('transmitBidActionBtn').innerText = `🚀  
TRANSMIT BID NOW (${value.toFixed(2)} ETH)`;  
    addLogEntry(`Bid amount scale mutated quickly via  
layout pad to: ${value.toFixed(2)} ETH`);  
    saveAppState();  
}
```

```
function handleManualBidInput(value) {  
    if (isCoinbaseDown) return;  
    let parsed = parseFloat(value);  
    if (!isNaN(parsed) && parsed > 0) {  
        activeBidValue = parsed;  
  
document.getElementById('activeBidDisplay').innerText =  
parsed.toFixed(2) + ' ETH';
```

```
document.getElementById('transmitBidActionBtn').innerText = `🚀  
TRANSMIT BID NOW (${parsed.toFixed(2)} ETH)`;  
    saveAppState();  
}
```

```

    }

    // INTERACTIVE BID TRANSMISSION SIMULATION DISPATCHER
    function executeBidTransmissionSim() {
        if (isCoinbaseDown) {
            addLogEntry("[DENIED] Aborting transaction
broadcast: Hardware circuit breaker is explicitly engaged.");
            return;
        }

        const platform =
document.getElementById('platformTypeSelect').value;

        const totalSeconds = 900;
        const remainingSeconds = totalSeconds - elapsedSeconds;

        const networkLagBuffer = 3.0;
        const totalRequiredLeadTime = userExecutionLag +
networkLagBuffer;

        addLogEntry(`[TRANSMIT] Broadcasting
${activeBidValue.toFixed(2)} ETH entry target position position
stance: [${userStance}]`);

        let isTxSuccessful = false;
        if (platform === 'prelocked') {
            if (remainingSeconds >= (60 + userExecutionLag)) {
                isTxSuccessful = true;
            }
        } else {
            if (remainingSeconds >= totalRequiredLeadTime) {
                isTxSuccessful = true;
            }
        }
    }

    totalTx++;
    const banner =
document.getElementById('optimalWindowBanner');
    if (isTxSuccessful) {
        successfulTx++;
        addLogEntry(`[SUCCESS] Tx Confirmed! Entry of

```

```

    ${activeBidValue.toFixed(2)} ETH on ${userStance} processed with
    ${remainingSeconds - totalRequiredLeadTime}.toFixed(1)}s safety
    window cushion remaining.`);
    banner.className = "p-2.5 rounded-xl border
    text-center text-xs font-mono font-bold bg-emerald-500
    text-slate-950 shadow-md shadow-emerald-500/20";
    banner.innerText = `✅ ORDER MATCHED:
    ${activeBidValue.toFixed(2)} ETH ON ${userStance}`;
    setTimeout(() => { syncTimerDisplay(); }, 2200);
  } else {
    failedTx++;
    addLogEntry(`[TX DROPPED] Late Submission! Total
    execution delay overhead (${totalRequiredLeadTime.toFixed(2)}s)
    exceeded the block gateway time capacity
    (${remainingSeconds.toFixed(1)}s remaining).`);
    banner.className = "p-2.5 rounded-xl border
    text-center text-xs font-mono font-bold bg-rose-600 text-white
    shadow-md shadow-rose-600/20 animate-bounce";
    banner.innerText = `❌ TX SLIPPED: GATEWAY
    TERMINATION TIMEOUT`;
    setTimeout(() => { syncTimerDisplay(); }, 2200);
  }
  updateTxLedgerDisplay();
  saveAppState();
}

```

```

function updateTxLedgerDisplay() {
  document.getElementById('txSuccess').innerText =
  successfulTx;
  document.getElementById('txFailed').innerText =
  failedTx;
  document.getElementById('txTotal').innerText = totalTx;
  const rate = totalTx > 0 ? (successfulTx / totalTx *
  100).toFixed(1) : '0.0';
  document.getElementById('txRateBadge').innerText = `TX
  RATE: ${rate}%`;
}

```

```

// BACKTESTING PROCESSING MATRIX COMPUTATION SUB-ENGINE
ROUTINE
function executeBulkBacktest() {
  const sampleSize =
  parseInt(document.getElementById('btSampleSize').value);
  const environmentBias =

```

```

document.getElementById('btMarketBias').value;
    const tableBody =
document.getElementById('btRegistryTableBody');

    tableBody.innerHTML = '';

    let btWins = 0;
    let btLosses = 0;


    let m10Wins = 0, m10Total = 0;
    let m85Wins = 0, m85Total = 0;
    let m375Wins = 0, m375Total = 0;


    let overGenerationProbability = 0.50;
    if (environmentBias === 'bullish')
overGenerationProbability = 0.65;
    if (environmentBias === 'bearish')
overGenerationProbability = 0.35;


    for (let b = 1; b <= sampleSize; b++) {
        let sampleSequence = [];
        for (let s = 0; s < 6; s++) {
            sampleSequence.push(Math.random() <
overGenerationProbability ? 1 : 0);
        }


        let totalWeight = 0;
        let overWeightedSum = 0;
        for (let i = 0; i < sampleSequence.length; i++) {
            let weight = 1.0 + (i * 0.15);
            totalWeight += weight;


            if (sampleSequence[i] === 1) overWeightedSum +=
weight;
        }
        const sampleBaseProbOver = overWeightedSum /
totalWeight;
        const sampleTotalOver = Math.max(0.01,
Math.min(0.99, sampleBaseProbOver + microMomentum));
        const sampleTotalUnder = 1.0 - sampleTotalOver;

```

```
const actualBlockResult = Math.random() <
overGenerationProbability ? 'OVER' : 'UNDER';
```

```
let overStanceSignal = sampleTotalOver >=
confidenceThreshold ? 'KEEP' : 'LET GO';
let underStanceSignal = sampleTotalUnder >=
confidenceThreshold ? 'KEEP' : 'LET GO';
```

```
let blockWinCount = 0;
let blockLossCount = 0;
```

```
const isOverCorrect = actualBlockResult === 'OVER';
if (overStanceSignal === 'KEEP') {
  if (isOverCorrect) { m10Wins++;
blockWinCount++; } else { blockLossCount++; }
} else {
  if (!isOverCorrect) { m10Wins++;
blockWinCount++; } else { blockLossCount++; }
```

```
}
m10Total++;
```

```
if (underStanceSignal === 'KEEP') {
  if (!isOverCorrect) { m85Wins++; m375Wins++;
blockWinCount += 2; } else { blockLossCount += 2; }
} else {
  if (isOverCorrect) { m85Wins++; m375Wins++;
blockWinCount += 2; } else { blockLossCount += 2; }
}
m85Total++;
m375Total++;
```

```
btWins += blockWinCount;
btLosses += blockLossCount;
```

```
const row = document.createElement('tr');
row.className = "hover:bg-slate-900/40 transition
duration-100";
```

```
const sequenceFormatted = sampleSequence.map(v => v
=== 1 ? '<span class="text-emerald-400">O</span>' : '<span
class="text-rose-400">U</span>').join('-');
const performanceBadgeClass = blockWinCount >=
blockLossCount ? 'text-emerald-400 font-bold' : 'text-rose-400';
```

```

        row.innerHTML = `
            <td class="p-3 text-slate-500
font-mono">#BT- $\{b.toString().padStart(3, '0')\}$ </td>
            <td class="p-3 font-mono
tracking-wider"> $\{\{sequenceFormatted\}\}$ </td>
            <td class="p-3 text-center"><span class="px-1
py-0.5 rounded text-[10px]  $\{\{overStanceSignal === 'KEEP' ?
'bg-emerald-500/10 text-emerald-400' : 'bg-slate-800
text-slate-500'\}$ > $\{\{overStanceSignal\}\}$ </span></td>
            <td class="p-3 text-center"><span class="px-1
py-0.5 rounded text-[10px]  $\{\{underStanceSignal === 'KEEP' ?
'bg-emerald-500/10 text-emerald-400' : 'bg-slate-800
text-slate-500'\}$ > $\{\{underStanceSignal\}\}$ </span></td>
            <td class="p-3 text-center"><span class="px-1
py-0.5 rounded text-[10px]  $\{\{underStanceSignal === 'KEEP' ?
'bg-emerald-500/10 text-emerald-400' : 'bg-slate-800
text-slate-500'\}$ > $\{\{underStanceSignal\}\}$ </span></td>
            <td class="p-3 font-bold  $\{\{actualBlockResult
=== 'OVER' ? 'text-emerald-400' :
'text-rose-400'\}$ > $\{\{actualBlockResult\}\}$ </td>
            <td class="p-3 text-right
 $\{\{performanceBadgeClass\}\}$ >+ $\{\{blockWinCount\}W$  /
- $\{\{blockLossCount\}L$ </td>
        `;
        tableBody.appendChild(row);
    }

    document.getElementById('btMetricTotal').innerText =
sampleSize;
    document.getElementById('btMetricWins').innerText =
btWins;
    document.getElementById('btMetricLosses').innerText =
btLosses;

    const aggregateSum = btWins + btLosses;
    const aggregateWinPercentage = (btWins / aggregateSum)
* 100;
    document.getElementById('btMetricWinRate').innerText =
aggregateWinPercentage.toFixed(1) + '%';

```

```

        document.getElementById('btHit-10m').innerText =
((m10Wins / m10Total) * 100).toFixed(1) + '%';
        document.getElementById('btHit-8_5m').innerText =
((m85Wins / m85Total) * 100).toFixed(1) + '%';
        document.getElementById('btHit-3_75m').innerText =
((m375Wins / m375Total) * 100).toFixed(1) + '%';

```

```

        const summaryStatus =
document.getElementById('btSummaryStatus');
        summaryStatus.className = "text-[10px] font-mono
font-bold uppercase tracking-wider px-2 py-0.5 bg-emerald-500
text-slate-950 rounded";
        summaryStatus.innerText = "MATRIX RUN SUCCESSFUL";

```

```

        addLogEntry(`[MATRIX SIMULATION] Successfully batched
and backtested ${sampleSize} active structural records
sequences.`);
    }

```

```

// TIME-DECAY ADVISOR MODULATED BY USER ACTION SPEED
LATENCY
        function updateOptimalWindowIndicator(remainingSeconds) {
            const banner =
document.getElementById('optimalWindowBanner');
            const platform =
document.getElementById('platformTypeSelect').value;

```

```

            const budgetValText =
document.getElementById('latencyBudgetVal');
            const budgetBar =
document.getElementById('latencyProgressBar');

```

```

            if (platform === 'prelocked') {
                if (remainingSeconds > 60) {
                    banner.className = "p-2.5 rounded-xl border
text-center text-xs font-mono font-bold transition-all duration-200
bg-slate-950 border-slate-800 text-slate-400 flex items-center
justify-center";
                    banner.innerText = "⌚ ROUND PREPARATION WINDOW
ACTIVE";

```

```

                    budgetValText.className = "text-emerald-400
font-bold";
                    budgetValText.innerText = "2.50s (SAFE)";

```

```

        budgetBar.className = "h-full bg-emerald-500
transition-all duration-300";
        budgetBar.style.width = "100%";
    } else if (remainingSeconds <= 60 &&
remainingSeconds >= 15) {
        banner.className = "p-2.5 rounded-xl border
text-center text-xs font-mono font-bold transition-all duration-200
bg-emerald-500/20 border-emerald-500/50 text-emerald-400
animate-pulse flex items-center justify-center";
        banner.innerText = "🔥 SUBMIT POSITION
IMMEDIATELY";

```

```

        let speedRatio = (remainingSeconds - 15) / 45;
        let currentBudget = (0.50 + (speedRatio *
2.00)).toFixed(2);

```

```

        budgetValText.className = "text-amber-400
font-bold";
        budgetValText.innerText = `${currentBudget}s
(TIGHT)`;
        budgetBar.className = "h-full bg-amber-500
transition-all duration-300";

```

```

        budgetBar.style.width = (speedRatio * 100) +
"%";
    } else {
        banner.className = "p-2.5 rounded-xl border
text-center text-xs font-mono font-bold transition-all duration-200
bg-rose-600/20 border-rose-500/40 text-rose-400 flex items-center
justify-center";
        banner.innerText = "🚫 BLOCK LOCK EXPIRED /
ARBITRARY REJECTION";

```

```

        budgetValText.className = "text-rose-500
font-bold animate-pulse";
        budgetValText.innerText = "0.00s (CRITICAL)";
        budgetBar.className = "h-full bg-rose-600
transition-all duration-300";
        budgetBar.style.width = "0%";
    }
} else {
    if (remainingSeconds > 60) {
        banner.className = "p-2.5 rounded-xl border
text-center text-xs font-mono font-bold transition-all duration-200
bg-slate-950 border-slate-800 text-slate-500 flex items-center

```

```

justify-center";
        banner.innerText = "⌚ ACCUMULATING DATA
EDGES";

        budgetValText.className = "text-emerald-400
font-bold";
        budgetValText.innerText = "3.00s (SAFE)";
        budgetBar.className = "h-full bg-emerald-500
transition-all duration-300";
        budgetBar.style.width = "100%";

    } else if (remainingSeconds <= 60 &&
remainingSeconds >= 15) {
        banner.className = "p-2.5 rounded-xl border
text-center text-xs font-mono font-bold transition-all duration-200
bg-emerald-500/20 border-emerald-500/50 text-emerald-400
animate-pulse flex items-center justify-center";
        banner.innerText = "🔥 OPTIMAL ENTRY SWEET
SPOT";

        let speedRatio = (remainingSeconds - 15) / 45;
        let currentBudget = (0.75 + (speedRatio *
2.25)).toFixed(2);

        budgetValText.className = "text-emerald-400
font-bold";
        budgetValText.innerText = `${currentBudget}s
(OPTIMAL)`;
        budgetBar.className = "h-full bg-emerald-500
transition-all duration-300";
        budgetBar.style.width = (speedRatio * 100) +
"%";
    } else {
        banner.className = "p-2.5 rounded-xl border
text-center text-xs font-mono font-bold transition-all duration-200
bg-amber-500/20 border-amber-500/40 text-amber-400 flex
items-center justify-center";
        banner.innerText = "⚠️ VOLATILITY LIQUIDITY
SQUEEZE CRUNCH";

        budgetValText.className = "text-rose-400
font-bold";
        budgetValText.innerText = "0.45s (LAG RISK)";
        budgetBar.className = "h-full bg-amber-600
transition-all duration-300";

```

```
budgetBar.style.width = "15%";
```

```
    }  
  }  
}
```

```
function initControlEventListeners() {  
  const tSlider =  
document.getElementById('thresholdSlider');  
  tSlider.addEventListener('input', (e) => {  
    confidenceThreshold = parseInt(e.target.value) /  
100;
```

```
document.getElementById('thresholdValueDisplay').innerText =  
e.target.value + '%';  
    executeCalculations();  
    saveAppState();  
  });
```

```
    const mSlider =  
document.getElementById('momentumSlider');  
    mSlider.addEventListener('input', (e) => {  
      microMomentum = parseInt(e.target.value) / 100;  
      const prefix = microMomentum >= 0 ? '+' : '';
```

```
document.getElementById('momentumValueDisplay').innerText = prefix  
+ e.target.value + '%';  
    executeCalculations();
```

```
    saveAppState();  
  });
```

```
    const tScrubber =  
document.getElementById('timeScrubber');  
    tScrubber.addEventListener('input', (e) => {  
      elapsedSeconds = parseInt(e.target.value);  
      if (isSimPlaying) togglePlayback();  
      syncTimerDisplay();  
      executeCalculations();  
    });  
  }
```

```

        function renderHistoricalBlocks() {
            const container =
document.getElementById('historicalBlocksGrid');
            container.innerHTML = '';

            for (let i = 0; i < historicalResults.length; i++) {
                const isOver = historicalResults[i] === 1;
                const positionLabel = i ===
historicalResults.length - 1 ? 'Newest' : (i === 0 ? 'Oldest' :
`T-${historicalResults.length - 1 - i}`);

                const blockDiv = document.createElement('div');
                blockDiv.className = `p-2 rounded-xl text-center
border cursor-pointer select-none transition ${
                    isOver
                        ? 'bg-emerald-500/10 border-emerald-500/30
text-emerald-400 hover:bg-emerald-500/20'
                        : 'bg-rose-500/10 border-rose-500/30
text-rose-400 hover:bg-rose-500/20'
                };
                blockDiv.innerHTML = `
                    <div class="text-[9px] font-mono text-slate-500
uppercase">${positionLabel}</div>
                    <div class="text-xs font-bold font-mono
my-0.5">${isOver ? 'OVER' : 'UNDER'}</div>
                `;
                blockDiv.onclick = () => toggleHistoricalBlock(i);
                container.appendChild(blockDiv);
            }

document.getElementById('historicalCountLabel').innerText =
`${historicalResults.length} Cycles Stacked`;
        }

        function toggleHistoricalBlock(index) {
            historicalResults[index] = historicalResults[index] ===
1 ? 0 : 1;

            renderHistoricalBlocks();
            addLogEntry(`Historical block index ${index + 1}
updated to ${historicalResults[index] === 1 ? 'OVER' : 'UNDER'}`);
        }

```

```

        executeCalculations();
        saveAppState();
    }

    function applyPreset(arrayData) {
        historicalResults = [...arrayData];
        renderHistoricalBlocks();
        addLogEntry(`Preset sequence profile applied:
[${arrayData.map(v => v === 1 ? 'O' : 'U').join(',')}]`);
        executeCalculations();
        saveAppState();
    }

    function setUserStance(stance) {
        userStance = stance;
        syncStanceUI();
        addLogEntry(`Target intention preference locked to:
${stance}`);
        executeCalculations();
    }

    saveAppState();
}

function syncStanceUI() {
    const overBtn =
document.getElementById('stanceOverBtn');
    const underBtn =
document.getElementById('stanceUnderBtn');

    if (userStance === 'OVER') {
        overBtn.className = "py-2 px-3 rounded-xl border
text-xs font-semibold tracking-wide font-mono transition flex
items-center justify-center gap-1.5 bg-emerald-500/10
border-emerald-500/40 text-emerald-400 shadow-sm
shadow-emerald-500/5";
        underBtn.className = "py-2 px-3 rounded-xl border
text-xs font-semibold tracking-wide font-mono transition flex
items-center justify-center gap-1.5 bg-slate-800/40
border-slate-700 text-slate-400";
    } else {
        underBtn.className = "py-2 px-3 rounded-xl border
text-xs font-semibold tracking-wide font-mono transition flex
items-center justify-center gap-1.5 bg-rose-500/10
border-rose-500/40 text-rose-400 shadow-sm shadow-rose-500/5";
        overBtn.className = "py-2 px-3 rounded-xl border

```

```
text-xs font-semibold tracking-wide font-mono transition flex
items-center justify-center gap-1.5 bg-slate-800/40
border-slate-700 text-slate-400";
    }
}
```

```
function executeCalculations() {
    if (historicalResults.length === 0) return;
```

```
    let totalWeight = 0;
    let overWeightedSum = 0;
```

```
    for (let i = 0; i < historicalResults.length; i++) {
        let weight = 1.0 + (i * 0.15);
        totalWeight += weight;
        if (historicalResults[i] === 1) overWeightedSum +=
weight;
    }
```

```
    const baseProbOver = overWeightedSum / totalWeight;
    const currentMomentumWeight = microMomentum;
```

```
    let finalProbOver = baseProbOver +
currentMomentumWeight;
    finalProbOver = Math.max(0.01, Math.min(0.99,
finalProbOver));
    const finalProbUnder = 1.0 - finalProbOver;
```

```
document.getElementById('overProbabilityDisplay').innerText =
(finalProbOver * 100).toFixed(1) + '%';
```

```
document.getElementById('underProbabilityDisplay').innerText =
(finalProbUnder * 100).toFixed(1) + '%';
```

```
    document.getElementById('overProbBarTop').style.width =
(finalProbOver * 100) + '%';
    document.getElementById('underProbBarTop').style.width
= (finalProbUnder * 100) + '%';
```

```

        document.getElementById('overBarSlider').style.width =
        (finalProbOver * 100) + '%';
        document.getElementById('underBarSlider').style.width =
        (finalProbUnder * 100) + '%';

```

```

        document.getElementById('overProbBreakdown').innerText
= `Base: ${ (baseProbOver * 100).toFixed(1)}% | Momentum Factor:
${ (currentMomentumWeight * 100).toFixed(1)}%`;
        document.getElementById('underProbBreakdown').innerText
= `Base: ${ ((1-baseProbOver) * 100).toFixed(1)}% | Momentum Factor:
${ (-currentMomentumWeight * 100).toFixed(1)}%`;

```

```

        evaluateRealtimeDirective(finalProbOver,
finalProbUnder);
        checkMilestoneTriggers(finalProbOver, finalProbUnder);
    }

```

```

        function evaluateRealtimeDirective(probOver, probUnder) {
            const dText = document.getElementById('directiveText');
            const dBadge =
document.getElementById('directiveStatusBadge');
            const dIcon =
document.getElementById('directiveIconBox');
            const dBox =
document.getElementById('activeDirectiveBox');

```

```

            let activeStanceProbability = userStance === 'OVER' ?
probOver : probUnder;
            let meetsThreshold = activeStanceProbability >=
confidenceThreshold;

```

```

            if (meetsThreshold) {
                dBadge.className = "w-full md:w-auto inline-block
text-center text-xl font-mono font-black uppercase tracking-wider
px-6 py-3 rounded-xl bg-gradient-to-r from-emerald-500 to-teal-600
text-slate-950 shadow-md shadow-emerald-500/20 animate-pulse
border-none";
                dBadge.innerText = "KEEP POSITION";
                dText.innerText = `CONFIRMED WINNING DECISION
METRICS: True statistical bias for [${userStance}] is currently
running at ${ (activeStanceProbability * 100).toFixed(1)}%, clearing
your strategy safety confidence threshold ruleset
${ (confidenceThreshold * 100).toFixed(0)}%`.`;
                dIcon.className = "h-14 w-14 rounded-2xl

```

```

bg-emerald-500/20 text-emerald-400 flex items-center justify-center
font-bold text-xl font-mono shrink-0";
        dIcon.innerText = "YES";
        dBox.className = "bg-slate-950 border-2
border-emerald-500/60 p-5 rounded-2xl flex flex-col md:flex-row
items-center justify-between gap-4 mt-1 transition-all duration-300
shadow-xl shadow-emerald-500/5";
    } else {
        dBadge.className = "w-full md:w-auto inline-block
text-center text-xl font-mono font-black uppercase tracking-wider
px-6 py-3 rounded-xl bg-gradient-to-r from-rose-600 to-red-700
text-white shadow-md shadow-rose-600/20 border-none";
        dBadge.innerText = "LET GO";
        dText.innerText = `HIGH PROBABILITY TEMPORAL RISK
DETECTED: Bias calculation strength for [{userStance}] dropped
down to ${ (activeStanceProbability * 100).toFixed(1)}%, failing to
sustain execution efficiency benchmarks (${ (confidenceThreshold *
100).toFixed(0)}%). Skip entry.`;
        dIcon.className = "h-14 w-14 rounded-2xl
bg-rose-500/20 text-rose-400 flex items-center justify-center
font-bold text-xl font-mono shrink-0";
        dIcon.innerText = "NO";
        dBox.className = "bg-slate-950 border-2
border-rose-600/40 p-5 rounded-2xl flex flex-col md:flex-row
items-center justify-between gap-4 mt-1 transition-all duration-300
shadow-xl shadow-rose-600/5";
    }
}
}

```

```

function checkMilestoneTriggers(probOver, probUnder) {
    let activeStanceProbability = userStance === 'OVER' ?
probOver : probUnder;

```

```

        Object.keys(MILESTONES).forEach(key => {
            const targetSeconds = MILESTONES[key];
            const card =
document.getElementById(`msCard-${key}`);
            const badge =
document.getElementById(`msBadge-${key}`);
            const stanceText =
document.getElementById(`msStance-${key}`);
            const probText =
document.getElementById(`msProb-${key}`);

```

```

        if (elapsedSeconds >= targetSeconds) {
          if (!lockedMilestoneData[key]) {
            let signalOutcome = activeStanceProbability
            >= confidenceThreshold ? 'KEEP' : 'LET GO';
            lockedMilestoneData[key] = {
              stance: userStance,
              prob: (activeStanceProbability *
100).toFixed(1) + '%',
              signal: signalOutcome
            };
            addLogEntry(`[MARK TRIGGERED] Captured
${key.replace('_', ' ')} mark logic snapshot.`);
          }
        }

```

```

        stanceText.innerText =
lockedMilestoneData[key].stance;
        probText.innerText =
lockedMilestoneData[key].prob;
        badge.innerText =
lockedMilestoneData[key].signal;

        if (lockedMilestoneData[key].signal === 'KEEP')
{
          badge.className = "text-[10px] font-mono
bg-emerald-500 text-slate-950 px-1.5 py-0.5 rounded font-bold
uppercase";
          card.className = "bg-slate-950 border
border-emerald-500/30 rounded-xl p-3 flex flex-col gap-2 transition
duration-200";
        } else {
          badge.className = "text-[10px] font-mono
bg-rose-600 text-white px-1.5 py-0.5 rounded font-bold uppercase";
          card.className = "bg-slate-950 border
border-rose-500/30 rounded-xl p-3 flex flex-col gap-2 transition
duration-200";
        }
      } else {
        if (lockedMilestoneData[key])
lockedMilestoneData[key] = null;
        badge.innerText = "PENDING";
        badge.className = "text-[10px] font-mono
bg-slate-900 text-slate-500 border border-slate-800 px-1.5 py-0.5
rounded font-bold uppercase";
        stanceText.innerText = "--";
      }
    }
  }

```

```
probText.innerText = "--";
```

```
        card.className = "bg-slate-950 border  
border-slate-800 rounded-xl p-3 flex flex-col gap-2 transition  
duration-200";  
    }  
    });  
}
```

```
function settleCurrentCycle(actualOutcome) {  
    let cycleWins = 0;  
    let cycleLosses = 0;  
    let capturedCheckpoints = 0;
```

```
    Object.keys(MILESTONES).forEach(key => {  
        const snapshot = lockedMilestoneData[key];  
        if (snapshot) {  
            capturedCheckpoints++;  
            const isStanceCorrect = snapshot.stance ===  
actualOutcome;
```

```
            if (snapshot.signal === 'KEEP') {  
                if (isStanceCorrect) cycleWins++; else  
cycleLosses++;  
            } else if (snapshot.signal === 'LET GO') {
```

```
                if (!isStanceCorrect) cycleWins++; else  
cycleLosses++;  
            }  
        }  
    });
```

```
    if (capturedCheckpoints === 0) {  
        addLogEntry("[LEDGER ERROR] Cannot settle. Advance  
time bar past a checkpoint first.");  
        return;  
    }
```

```
    totalWins += cycleWins;  
    totalLosses += cycleLosses;
```

```
    document.getElementById('ledgerWins').innerText =
```

```

totalWins;
    document.getElementById('ledgerLosses').innerText =
totalLosses;

    const totalElements = totalWins + totalLosses;
    document.getElementById('ledgerTotal').innerText =
totalElements;

    const numericalWinRate = (totalWins / totalElements) *
100;
    document.getElementById('winRateBadge').innerText =
`WR: ${numericalWinRate.toFixed(1)}%`;

    addLogEntry(`[LEDGER RESOLVED] Candle actual settled:
${actualOutcome}. Recorded: +${cycleWins}W / -${cycleLosses}L.`);
    resetPlayback();
    saveAppState();
}

function startTickerTapeStream() {
    const tape =
document.getElementById('tickerTapeStream');
    setInterval(() => {
        if (isSimPlaying || timerMode === 'live') {
            const isBuy = Math.random() > (0.50 -
(microMomentum * 0.3));
            const volume = (Math.random() * 4.8 +
0.05).toFixed(3);
            const costDelta = (Math.random() * 24 +
1).toFixed(2);
            const timestamp = new
Date().toLocaleTimeString().split(' ')[0];

            const item = document.createElement('div');
            if (isBuy) {

                item.className = "text-emerald-400 flex
justify-between items-center border-b border-slate-900 pb-1";
                item.innerHTML = `[${timestamp}]
INFLOW</span> <span class="font-bold font-mono">+${volume}
ETH</span> <span>+${costDelta}</span>`;

```

```

        } else {
            item.className = "text-rose-400 flex
justify-between items-center border-b border-slate-900 pb-1";
            item.innerHTML = `[${timestamp}]
OUTFLOW</span> <span class="font-bold font-mono">-${volume}
ETH</span> <span>-${costDelta}</span>`;
        }
        tape.insertBefore(item, tape.firstChild);
        if (tape.children.length > 20)
tape.removeChild(tape.lastChild);
    }
}, 350);
}

```

```

function syncTimerDisplay() {
    document.getElementById('timeScrubber').value =
elapsedSeconds;
    let totalSeconds = 900;
    let remainingSeconds = totalSeconds - elapsedSeconds;

    let rMin = Math.floor(remainingSeconds / 60);
    let rSec = remainingSeconds % 60;
}

```

```

document.getElementById('countdownTimerDisplay').innerText =
`${rMin.toString().padStart(2,
'0')}:${rSec.toString().padStart(2, '0')}`;

```

```

document.getElementById('elapsedSecondsDisplay').innerText =
`${elapsedSeconds}s elapsed of ${totalSeconds}s`;

```

```

    updateOptimalWindowIndicator(remainingSeconds);
}

```

```

function togglePlayback() {
    const btn = document.getElementById('playPauseBtn');
    if (isSimPlaying) {
        clearInterval(simIntervalId);
        isSimPlaying = false;
        btn.innerHTML = `

```

```

    } else {
        if (elapsedSeconds >= 900) elapsedSeconds = 0;
        isSimPlaying = true;

        btn.innerHTML = `<svg class="w-3.5 h-3.5
fill-current" viewBox="0 0 24 24"><path d="M6
19h4V5H6v14zm8-14v14h4V5h-4z"/></svg> Pause Sim`;
        addLogEntry("Probability logic ticker engine
active.");

        simIntervalId = setInterval(() => {
            elapsedSeconds += 1;
            if (elapsedSeconds > 900) {
                clearInterval(simIntervalId);
                isSimPlaying = false;
                btn.innerHTML = `<svg class="w-3.5 h-3.5
fill-current" viewBox="0 0 24 24"><path d="M8 5v14l11-7z"/></svg>
Play Sim`;
                addLogEntry("[CYCLE FINISHED] End of 15
minute interval block reached.");
            }
            syncTimerDisplay();
            executeCalculations();
        }, 100);
    }
}

function resetPlayback() {
    if (isSimPlaying) togglePlayback();

    elapsedSeconds = 0;
    syncTimerDisplay();
    executeCalculations();
    addLogEntry("Timeline workspace counters completely
cleared back to zero starting grid context.");
}

function jumpToSeconds(secondsValue) {
    elapsedSeconds = secondsValue;
    syncTimerDisplay();
    executeCalculations();
    addLogEntry(`Jumped timeline tracking point reference

```

```
to exactly: ${Math.floor((900-secondsValue)/60)}m  
${(900-secondsValue)%60}s remaining.`);  
}
```

```
function startSystemClock() {  
  setInterval(() => {  
    const now = new Date();  
    document.getElementById('systemClock').innerText =  
now.toISOString().replace('T', ' ').substring(0, 19) + ' UTC';  
  }, 1000);  
}
```

```
function addLogEntry(msg) {  
  const consoleBox =  
document.getElementById('logConsole');  
  const stamp = new Date().toLocaleTimeString();
```

```
  const logRow = document.createElement('div');  
  logRow.className = "border-b border-slate-900/60 pb-1";  
  logRow.innerHTML = `[${stamp}] <span  
class="text-slate-300">${msg}</span>`;
```

```
  consoleBox.appendChild(logRow);  
  consoleBox.scrollTop = consoleBox.scrollHeight;  
}
```

```
function clearLogs() {  
  document.getElementById('logConsole').innerHTML = '<div  
class="text-slate-500">[00:00:00] Probability environment console  
reset.</div>';  
}
```

```
function setTimerMode(mode) {  
  timerMode = mode;  
  const simBtn = document.getElementById('modeSimBtn');
```

```
  const liveBtn = document.getElementById('modeLiveBtn');
```

```
  if (mode === 'sim') {  
    simBtn.className = "px-2 py-1 rounded bg-slate-800
```

```
text-white font-medium";
        liveBtn.className = "px-2 py-1 rounded
text-slate-400 hover:text-slate-200";
        addLogEntry("Control system configured to
Accelerated Simulation Mode.");
    } else {
        simBtn.className = "px-2 py-1 rounded
text-slate-400 hover:text-slate-200";
        liveBtn.className = "px-2 py-1 rounded bg-slate-800
text-white font-medium";
        if (isSimPlaying) togglePlayback();
        addLogEntry("System attached to Live Cold Standby
mode.");
    }
}
</script>
</body>
</html>
```